



ENGINEERING MATHEMATICS - II

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ENGINEERING MATHEMATICS - II

UNIT 5 : Fourier series

Session : 10

Sub Topic : Problems on Complex Form of the Fourier Series

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Complex form of the Fourier series - Problems

1. Find the complex exponential form of the Fourier Series of $f(x) = e^{-x}$, $-1 < x < 1$

Solution:

The complex exponential form of the Fourier series is given by

$$f(x) = \sum_{n=-\infty}^{\infty} c_n e^{\frac{in\pi x}{l}}$$

where

$$c_n = \frac{1}{2l} \int_{-l}^l f(x) e^{\frac{-in\pi x}{l}} dx.$$

Here $l = 1$

Therefore,

$$f(x) = \sum_{n=-\infty}^{\infty} c_n e^{in\pi x}$$

Complex form of the Fourier series - Problems

$$c_n = \frac{1}{2} \int_{-1}^1 e^{-x} e^{\frac{-in\pi x}{1}} dx$$

$$c_n = \frac{1}{2} \int_{-1}^1 e^{-(1+in\pi)x} dx$$

$$= \frac{1}{2} \left[\left(\frac{e^{-(1+in\pi)x}}{-(1+in\pi)} \right)_{-1}^1 \right]$$

$$= \frac{1}{2} \left[\frac{e^{-(1+in\pi)x}}{-(1+in\pi)} - \frac{e^{(1+in\pi)x}}{-(1+in\pi)} \right]$$

$$c_n = \frac{1}{2(1+in\pi)} [e e^{in\pi} - e^{-1} e^{in\pi}]$$

Complex form of the Fourier series - Problems

$$\begin{aligned}c_n &= \frac{1}{2(1 + in\pi)} [e(\cos n\pi + i\sin n\pi) - e^{-1}(\cos n\pi - i\sin n\pi)] \\&= \frac{1}{2(1 + in\pi)} (e - e^{-1})(-1)^n \\&= \frac{(1 - in\pi)}{(1 + n^2\pi^2)2} (e - e^{-1})(-1)^n \\c_n &= \frac{(-1)^n(1 - in\pi)\sinh 1}{(1 + n^2\pi^2)}\end{aligned}$$

$$f(x) = e^{-x} = \sum_{-\infty}^{\infty} \frac{(-1)^n(1 - in\pi)\sinh 1}{(1 + n^2\pi^2)} e^{in\pi x}$$

This is the required complex exponential Fourier series

Complex form of the Fourier series - Problems

2. Obtain the complex form of the Fourier series of the function

$$f(x) = \begin{cases} 0 & -\pi \leq x < 0 \\ 1 & 0 < x \leq \pi \end{cases} \text{ and express it as real Fourier series.}$$

Solution

The complex exponential form of the Fourier series is given by

$$f(x) = \sum_{n=-\infty}^{\infty} c_n e^{inx} \dots\dots\dots(1)$$

where

$$c_n = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x) e^{-inx} dx$$

$$c_n = \frac{1}{2\pi} \left[\int_{-\pi}^0 0 \cdot e^{\frac{-in\pi x}{\pi}} dx + \int_0^{\pi} 1 \cdot e^{\frac{-in\pi x}{\pi}} dx \right]$$

Complex form of the Fourier series - Problems

$$= \frac{1}{2\pi} \int_0^{\pi} e^{-inx} dx$$

$$= \frac{1}{2\pi} \left[\frac{e^{-inx}}{-in} \right]_0^{\pi}$$

$$= \frac{1}{2\pi in} [-e^{-in\pi} + 1]$$

$$= \frac{1}{2\pi in} \left[1 - \overset{e^{-in\pi}}{\uparrow} (\cos n\pi - i \sin n\pi) \right]$$

$$C_n = \frac{1}{2\pi in} [1 - e^{-in\pi}] \text{ is valid when } n \neq 0$$

$$\begin{aligned} 1 - e^{-in\pi} &= 0 \\ 1 - (-1)^n &= 2 \end{aligned}$$

Complex form of the Fourier series - Problems

When $n=0$

$$C_0 = \frac{1}{2\pi} \int_0^{\pi} 1 \cdot dx$$

$$= \frac{1}{2\pi} [x]_0^{\pi}$$

$$= \frac{1}{2\pi} [\pi]$$

$$\boxed{C_0 = \frac{1}{2}}$$

$$C_n = \begin{cases} 0 & \text{when } n \text{ is even} \\ \frac{1}{\pi i n} & \text{when } n \text{ is odd} \end{cases}$$



From ①

$$f(x) = \sum_{n=-\infty}^{\infty} C_n e^{inx}$$

$$= \sum_{n=-\infty}^{-1} C_n e^{inx} + C_0 + \sum_{n=1}^{\infty} C_n e^{inx}$$

$$f(x) = \sum_{n=-\infty}^{-1} \frac{1}{\pi i n} e^{inx} + \frac{1}{2} + \sum_{n=1}^{\infty} \frac{1}{\pi i n} e^{inx}$$

$$= \frac{1}{2} + \frac{1}{i\pi} \left[\sum_{n=-\infty}^{-1} \frac{e^{inx}}{n} + \sum_{n=1}^{\infty} \frac{e^{inx}}{n} \right]$$

is the required complex exponential Fourier series

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Complex form of the Fourier series - Problems

$$\sin nx = \frac{e^{inx} - e^{-inx}}{2i}$$

$$= \frac{1}{2} + \frac{1}{i\pi} \left[\left(e^{ix} + \frac{e^{i3x}}{3} + \frac{e^{i5x}}{5} + \dots \right) + \left(\frac{e^{-ix}}{-1} + \frac{e^{-i3x}}{-3} + \frac{e^{-i5x}}{-5} + \dots \right) \right]$$

$$= \frac{1}{2} + \frac{1}{i\pi} \left[(e^{ix} - e^{-ix}) + \frac{1}{3} (e^{i3x} - e^{-i3x}) + \frac{1}{5} (e^{i5x} - e^{-i5x}) + \dots \right]$$

$$= \frac{1}{2} + \frac{2}{\pi} \left[\frac{e^{ix} - e^{-ix}}{2i} + \frac{1}{3} \left(\frac{e^{i3x} - e^{-i3x}}{2i} \right) + \frac{1}{5} \left(\frac{e^{i5x} - e^{-i5x}}{2i} \right) + \dots \right]$$

$$f(x) = \frac{1}{2} + \frac{2}{\pi} \left[\sin x + \frac{1}{3} \sin 3x + \frac{1}{5} \sin 5x + \dots \right]$$



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THANK YOU

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